# Anatomy of Runtime Symbolization Cost in Zephyr Firmware: A Pre-Registered Decomposition and Its Negative Outcome

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## Abstract

Zephyr's `CONFIG_SYMTAB` embeds a table of function addresses and name pointers so that fault handlers can print symbol names. Compact encodings of such tables (relative offsets, delta blocks, dictionary or front-coded names) are known from Linux `kallsyms` and from string compressors such as FSST. We ask a decomposition question: of the flash saved by a compact symbol table over the best existing scheme, how much comes from the *address index* representation and how much from *name coding*, once baselines receive the same compact address index? We froze a protocol with gates (G1–G5) and termination rules (T1–T3), with SHA-256 hashes, before measuring. We relinked ten existing Zephyr sample applications (six Cortex-M3, four RV32; 137–2,337 symbols; GCC 14.3) to a fixed point under 14 symbol-table schemes, and validated decoders against an oracle on the host and in QEMU. Compact schemes save 1.7–3.1% of the flash image (408 B–8.4 KB) over the best existing scheme; about 71% of that is attributable to the address index (pooled, exploratory metric). After equalizing the address index, the front-coded dictionary adds 0.4–0.8% for $N \geq 710$ and none at $N = 137$. **The pre-registered gates were not met**: the 70%-share gate (G1) and the FSST-within-1% gate (G3) failed, one baseline failed to reach a fixed point on one workload (T3), and the effect sizes sit at the thresholds. We therefore report a negative, observational result and make no claim of a successful contribution. Measurements are flash sizes only; no timing or energy claims are made.

## 1 Introduction

Firmware images that can print symbolized backtraces must carry a symbol table. Zephyr's stock table stores, per function, a 32-bit offset and a name pointer, searched by binary search [5]. Linux reduced its analogous table using relative offsets [1] and has seen proposals for delta-compressed tables [3]; FSST [2] is a general short-string compressor with random access. Partitioned Elias–Fano [4] is related prior work on compressed monotone sequences (not used in our experiments).

A compact table changes two things at once: how *addresses* are indexed and how *names* are stored. Reporting only the total saving conflates them, and comparing against a baseline lacking the compact address index overstates the value of name coding. This report isolates the two effects with controls, under a protocol frozen before measurement.

## 2 Method

**Schemes.** 14 arms per workload: existing (KALL256, KALL16, CAND_DIRECT, CAND_CP16), address-hardened controls (B1/B2 delta blocks × pair-token or kallsyms-token names), raw 32-bit address controls, the front-coded dictionary (`B1_B16_FC16`, `B2_B16_FC16`), a reference FSST arm (`B1_B16_FSST`), the stock Zephyr table, and no table.

**Fixed-point relinking.** The table contents depend on addresses, which depend on table size. For each arm we encode the manifest of $\mathrm{ELF}_{k-1}$, relink with the build's own link command, re-extract the manifest, and stop when equal (max. 8 iterations).

**Metrics.** Image size = flat binary (`objcopy -gap-fill 0xFF`). With $E$ the best existing arm, $H$ the best hardened control, $P$ the proposed arm: $S_{\mathrm{tot}} = E - P$, $S_{\mathrm{addr}} = E - H$, share $= S_{\mathrm{addr}}/S_{\mathrm{tot}}$, residual $g = (H - P)/H$.

**Gates.** G1: share $\geq 70\%$ on $\geq 75\%$ of workloads. G2: $g \leq 1\%$ on $\geq 75\%$. G3: FSST within 1% of $P$ for $N > 1000$. G4: zero decoder mismatches. G5: valid baseline and link checks. Terminate if T1 ($g > 1\%$ on half the workloads), T2 (share $< 50\%$) or T3 (a required arm fails to reach a fixed point).

**Validation.** Every configuration's decoder was checked against an oracle on host-compiled C; 8 configurations were also run on target in QEMU (Cortex-M3 `mps2-an385`, RV32 `virt`) via a self-test hashing every lookup result.

**Workloads.** Selected by a deterministic rule from a feasibility sweep (59 ARM samples built, $N$ up to 2,337)

*before* any size result was seen; the RV32 sweep was truncated at 20 samples for time. Frozen protocol: `GAP_FILL_PROTOCOL.md`.

# 3 Preliminary Measurement Blockers

**A.** A historical baseline reported a 21.5 MB final heap, another 1.65 MB, on the same trace. Replay shows two different classes share one name, and one harness additionally keeps a 20.2 MB unique-name set inside the measured scope. Neither is a defensible baseline, so all earlier memory comparisons are excluded here. **B.** Earlier "stock" images were only 264 B above no-table images because the harness never linked the generated table, only a 16-byte stub. The new harness reads the linked table back from the ELF and requires entry-by-entry equality (10/10 workloads).

# 4 Results

Table 1 gives per-workload sizes under the exploratory metric (sum of allocated PROGBITS section sizes). The frozen flat-binary metric quantizes RV32 images in exact 4096-B steps, which distorts small differences there; we kept the frozen metric as primary and added the exploratory metric, labelled as such.

**Total saving.** 1.7–3.1% of the image over the best existing scheme.

**Address share.** Pooled 71.2% (median 70.2%); five of nine usable shares lie within ±1 point of 70%, so the 75%-of-workloads rule is neither confidently met nor refuted.

**Residual name coding.** 0.43–0.83% for $N \geq 710$, $-0.64\%$ at $N = 137$.

**FSST.** 0.5–1.2% larger than $P$ for $N > 1000$ and $+7.2\%$ at $N = 137$.

**Decoders.** Zero mismatches on every converged arm; 8/8 QEMU self-tests passed (lookup correctness only).

# 5 Discussion and Limitations

The pre-registered gates were not met (Table 2), so the direction is stopped. KALL16 on `net_ptp` oscillated between 242,152 and 242,160 B; the cause was not investigated, and 139 of 140 fixed points converged in 2–3 iterations. On RV32 the stock lookup function (96 B) remains linked in every candidate arm, failing the frozen stub check; it cancels in $E/H/P$ comparisons. Even excluding `net_ptp`, G1 and G3 fail, and the effect sizes lie at the thresholds; adjusting them post hoc is forbidden by the protocol. The mechanisms (relative offsets, block deltas, FSST) have close prior art, and the remaining finding is an engineering decomposition of 1–3% of an image.

*Limitations:* no sample reached ≈3,000 symbols (max 2,337); one compiler family in the gate set; two architectures; flash only, no timing, energy or RAM-buffer claims; all checks by the project's own tooling, not independently verified. The "Cromie" patch mentioned in an earlier audit could not be located and is not cited.

*Disclosure and authorship:* repository patent notes document public-disclosure risk; whether filing or release is intended is unresolved. Author agreement is not recorded, so no author list is given.

# 6 Reproducibility

All scripts, frozen protocol, hashes (`FREEZE_MANIFEST.txt`), raw per-workload results, stripped ELFs, link maps and generated C are under `research/paper_final/`; see `ARTIFACT_MAP.md` and `CLAIMS_LEDGER.md`. Requires Zephyr SDK 1.0.1, Zephyr @403a7d79, FSST @e638d4cf.

# References


[1] Ard Biesheuvel. kallsyms: add support for relative offsets in kallsyms address table. Linux kernel commit 2213e9a66bb87d8344a1256b4ef568220d9587fb, 2016. Committed by Linus Torvalds, March 2016.

[2] Peter Boncz, Thomas Neumann, and Viktor Leis. FSST: Fast random access string compression. *Proceedings of the VLDB Endowment*, 13(12):2649–2661, 2020.

[3] Sasha Levin. kallsyms: delta-compress lineinfo tables for 2.7x size reduction (PATCH v7 3/4). Linux kernel mailing list, patch series (v2 March 2026 – v7 July 2026); mirror: https://lkml.iu.edu/2607.1/04652.html, 2026. Under review at time of access; not known to be merged.

[4] Giuseppe Ottaviano and Rossano Venturini. Partitioned Elias–Fano indexes. In *Proceedings of the 37th International ACM SIGIR Conference on Research and Development in Information Retrieval*, Gold Coast, Australia, 2014.

[5] Zephyr Project. Zephyr RTOS source tree: `scripts/build/gen_symtab.py` and `subsys/debug/symtab/symtab.c`. Local checkout, commit 403a7d79adbce610cb611912190ba066d7c8903f (version 4.4.99), 2026.

| workload | arch | $N$ | $E$ | $H$ | $P$ | $S_{\text{tot}}$ | $S_{\text{addr}}$ | share | $g$ (%) |
|---|---|---|---|---|---|---|---|---|---|
| hello_world_rv32 | RV32 | 137 | 24394 | 23834 | 23986 | 408 | 560 | 137% | -0.64 |
| net_cloud_mqtt_azure_rv32 | RV32 | 2148 | 388818 | 383460 | 381100 | 7718 | 5358 | 69% | 0.62 |
| net_ethernet_bridge_rv32 | RV32 | 1445 | 281698 | 278182 | 276978 | 4720 | 3516 | 74% | 0.43 |
| net_ethernet_lldp_mps2 | M3 | 1422 | 225848 | 222332 | 220868 | 4980 | 3516 | 71% | 0.66 |
| net_midi2_rv32 | RV32 | 985 | 168980 | 166568 | 165434 | 3546 | 2412 | 68% | 0.68 |
| net_prometheus_mps2 | M3 | 2337 | 390776 | 384848 | 382336 | 8440 | 5928 | 70% | 0.65 |
| net_sockets_http_get_mps2 | M3 | 1061 | 141116 | 138472 | 137348 | 3768 | 2644 | 70% | 0.81 |
| net_sockets_socketpair_mps2 | M3 | 710 | 83912 | 82028 | 81344 | 2568 | 1884 | 73% | 0.83 |
| net_stats_mps2 | M3 | 1321 | 206932 | 203700 | 202308 | 4624 | 3232 | 70% | 0.68 |

Table 1: Bytes of flash image per scheme class (exploratory metric). $E$: best existing; $H$: best hardened control; $P$: front-coded dictionary. $g$: residual name-coding gain. `net_ptp` omitted (non-converged baseline).

| Gate | Frozen metric | Exploratory |
|---|---|---|
| G1 share (need 8/10) | 5 — fail | 6 — fail |
| G2 residual (need 8/10) | 7 — fail | 9 — pass |
| G3 FSST | 4/6 — fail | 5/6 — fail |
| G4 mismatches | fail* | fail* |
| G5 validity | fail | fail |
| T1 | not triggered | not triggered |
| T2 | not triggered | not triggered |
| T3 | **triggered** | **triggered** |

Table 2: Gate outcomes. *All mismatches arise from the non-converged KALL16 arm on `net_ptp`; zero on converged arms.